

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Mathematics — Class IX

Syllabus: Unit 1 — Real Numbers | Unit 2 — Logarithms | Unit 4 — Factorization & Algebraic Manipulation | Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

General Instructions:

- Write your name, roll number, and section clearly on the answer sheet.
- Section A** (MCQs): Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
- Section B** (Short Questions): Attempt **ALL 11** pairs, choosing the main question **OR** its alternative. Each carries **3 marks**.
- Section C** (Long Questions): Attempt **ALL 4** pairs, choosing the main question **OR** its alternative. Each carries **5 marks**.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Encircle / bubble the correct option. All questions are compulsory.

#	Question	A	B	C	D	Answer
1	If $a = b$, the property that lets us write $b = a$ is called the:	Reflexive property	Symmetric property	Transitive property	Additive property	○○○○
2	If $ac < bc$ and $c > 0$, then $a < b$ follows by the:	Trichotomy property	Cancellation (mult.) property	Additive property	Transitive property	○○○○
3	The scientific notation of 0.0000728 is:	7.28×10^{-5}	7.28×10^5	0.728×10^{-4}	72.8×10^{-6}	○○○○
4	Given $\log 2 = 0.3010$, the number of digits in 2^{60} is:	17	18	19	20	○○○○
5	Antilog of 3.0000 is:	3	100	1000	0.001	○○○○
6	The factors of $x^2 + 2x - 15$ are:	$(x + 5)(x - 3)$	$(x - 5)(x + 3)$	$(x + 15)(x - 1)$	$(x - 15)(x + 1)$	○○○○
7	The factorization of $8 + y^3$ is:	$(2 + y)(4 - 2y + y^2)$	$(2 + y)(4 + 2y + y^2)$	$(2 - y)(4 + 2y + y^2)$	$(y + 2)^3$	○○○○
8	HCF of P, Q is $(x - 3)$; their LCM is $(x - 3)(x + 2)(x - 5)$. The product $P \times Q$ is:	$(x - 3)^2(x + 2)(x - 5)$	$(x - 3)(x + 2)(x - 5)$	$(x - 3)^2(x + 2)$	$(x + 2)(x - 5)$	○○○○
9	The solution set of $ 6 - 3y = 0$ is:	{2}	{-2}	{0}	{6, -6}	○○○○
10	The solution of $-4x \geq 20$, $x \in \mathbb{R}$, is:	$x \leq -5$	$x \geq -5$	$x \leq 5$	$x \geq 5$	○○○○
11	The point $(-4, -(-6))$ lies in the:	First quadrant	Second quadrant	Third quadrant	Fourth quadrant	○○○○
12	If $x < 0$ and $y > 0$, then the point $(-x, y)$ lies in the:	First quadrant	Second quadrant	Third quadrant	Fourth quadrant	○○○○

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt ALL 11 questions. Choose either the main question OR its alternative.

Q.	Question	Marks	OR	OR Question	Marks
2(i)	Qasim had a balance of Rs. 1,75,000 in his account. He wrote one cheque for Rs. 45,790 and another for Rs. 1,12,921. Find his current balance.	3	OR	Salma bought 3.2 yards of fabric for a total price of Rs. 139.2. How much did the fabric cost per yard?	3
2(ii)	For a class party the students' council purchased 42 balloons at Rs. 1.85 each. Find the total amount paid for the balloons.	3	OR	A rectangular banner is 4 yards long and has a perimeter of 10 yards. (a) Find its width. (b) Find the cost of fabric for the whole perimeter at Rs. 3.75 per yard.	3
2(iii)	A hiking club went on a 7-day trip, hiking between 5.5 and 7.5 miles each day. Find the minimum and maximum possible total distance, and state which is reasonable: (a) less than 35 miles (b) between 35 and 55 miles (c) exactly 55 miles (d) more than 55 miles.	3	OR	Wajid drove 283.4 km using 16.2 litres of petrol and claims he averaged about 17.5 km/litre. Calculate his actual average and state whether his claim is reasonable.	3
2(iv)	Find the logarithm of: (a) 5313 (b) 1109, showing the characteristic and mantissa for each.	3	OR	Find the antilog of: (a) 3.5636 (b) 2.9281.	3
2(v)	It takes 5 seconds to recite 'Kalma Pak' once. How many hours will it take to recite it 10^6 (one million) times? Convert the answer into days (round down) and express in standard form.	3	OR	The distance between Earth and Sun is about 9.3×10^7 miles. If light travels at about 1.86×10^5 miles/second, find the time for light to reach Earth. Convert the answer to minutes, in standard form.	3
2(vi)	Find the square root of $16y^2 - 56y + 49$ by	3	OR	Factorize completely: $2x^2 - 5x - 3$.	3

	factorization.				
2(vii)	Find the HCF of $x^2 - 2x - 8$ and $x^2 - x - 12$ by factorization.	3	OR	Find the HCF of $2x^2 + 5x - 3$ and $x^2 + 2x - 3$ by factorization.	3
2(viii)	Solve the absolute value equation: $ z + 3 - 3 = 5 - z + 3 $.	3	OR	Solve the absolute value equation: $3 z - 2 - 4 = -2$.	3
2(ix)	Solve the compound inequality: $3x + 21 < 1 - x$ or $3x + 8 > 3 - 2x$, $x \in \mathbb{R}$.	3	OR	Write each in set-builder notation: (a) $2 \leq x < 5$, $x \in \mathbb{N}$ (b) $x \leq 4$, $x \in \mathbb{W}$ (c) $-4 < x \leq 3$, $x \in \mathbb{R}$.	3
2(x)	Square OLMN has O at the origin and L(3, 0) on the x-axis, with N on the y-axis and M diagonally opposite O. Find the coordinates of vertices M and N.	3	OR	Rectangle ABCD has vertex B(0, 2) on the y-axis and vertex D(-4, 0) on the x-axis, with A at the origin. Find the coordinates of vertices A and C.	3
2(xi)	Points A(3, 4) and C(x, y) are such that B(-1, 7) is the midpoint of AC. Find the coordinates of C.	3	OR	Triangle XYZ has vertices X(7, 4), Y(7, 1), Z(-3, 1). Using the distance formula, find the length of the hypotenuse and state which side it is.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Attempt ALL 4 questions. Choose either the main question OR its alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	Given $\log 2 = 0.3010$, $\log 3 = 0.4771$, $\log 5 = 0.6990$, $\log 7 = 0.8451$: (a) Evaluate $\log(63/20)$ using the laws of logarithms, showing each step. [3] (b) Find the number of digits in 6^{45} . [2]	5	OR	Given the same log values: (a) Evaluate $\log(84/25)$ using the laws of logarithms, showing each step. [3] (b) If $\log x = 2.4771$, find the value of x. [2]	5
3(ii)	The area of a rectangular veranda is given by $(x^2 - 2x - 3) \text{ m}^2$. Find: (a) expressions for both dimensions of the veranda [2] (b) the expression for its perimeter [1] (c) the cost of fencing it at Rs. 200/m [1] (d) the cost of carpeting the floor at Rs. 250/m ² [1]	5	OR	The area of a square machine surface is given by $(25x^2 - 30x + 9) \text{ m}^2$. Find: (a) the expression for the side length [2] (b) the expression for the boundary (perimeter) [1] (c) the cost of polishing at Rs. 75/m ² [1] (d) the cost of edging around 2 sides at Rs. 28/m [1]	5
3(iii)	(a) Solve the double inequality $1 \leq 5 - 3x \leq 22$, $x \in \mathbb{R}$, and write the solution set. Show it on a number line. [3] (b) Solve the absolute value equation $ 5y = 9$. [2]	5	OR	(a) Solve the compound inequality: $5 - 3x < 11$ or $2x + 3 < -9$, $x \in \mathbb{R}$. Write the solution set. [3] (b) Solve the radical equation $5 - \sqrt{2x - 1} = 0$, stating any restriction on x. [2]	5
3(iv)	Hassan and Ali start a race from different points on the coordinate axes, each 6 units from the origin — Hassan on the x-axis and Ali on the y-axis. They both run to a common target point C, which lies on the segment joining their starting points. (a) State the coordinates of Hassan's and Ali's starting points. [1] (b) If C is the midpoint of the segment joining them, find C. [2] (c) Verify using the distance formula that Hassan's distance to C equals Ali's distance to C. [2]	5	OR	Show that A(5, 8), B(7, 5), C(5, 2) and D(3, 5) are the vertices of a rhombus by finding the lengths of all four sides using the distance formula. [3] Then find the lengths of diagonals AC and BD. [2]	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Unit 1 (Real Numbers) | Unit 2 (Logarithms) | Unit 4 (Factorization & Algebraic Manipulation) | Unit 5 (Linear Equations & Inequalities) | Unit 7 (Coordinate Geometry)

— End of Question Paper —